

Independence and discrete random variables

Events · Variables · Counts

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Monday: independence and random variables

From Week 1

$$P(A \cap B) = P(A \mid B) P(B), \quad P(B) > 0.$$

Conditioning can change a probability.

Today

- ① Decide when it does not change.
- ② Assign numbers to outcomes.
- ③ Translate numerical statements into events.

Check the conditioning

Draw two cards without replacement from a standard deck. If the first is an ace, what is the chance the second is an ace?

Independent events

Definition

Events A and B are independent if

$$P(A \cap B) = P(A) P(B).$$

If $P(B) > 0$, this is equivalent to $P(A | B) = P(A)$.

Derive the equivalence from the conditional-probability formula.

The product definition still applies when one of the probabilities is zero.

Disjoint and independent are different conditions

Disjoint

$$A \cap B = \emptyset.$$

The events cannot occur together.

Independent

$$P(A \cap B) = P(A) P(B).$$

Their joint probability factors.

Suppose $P(A) > 0$ **and** $P(B) > 0$. Why can disjoint events not be independent?

If $P(A) = 0$, then A is independent of every event B : both sides are zero.

A random variable assigns a number to each outcome

Definition

A random variable is a real-valued function on the sample space:

$$X : \Omega \longrightarrow \mathbb{R}, \quad \omega \longmapsto X(\omega).$$

Example: one coin toss

$$\Omega = \{H, T\}.$$

$$H \longrightarrow \boxed{1}$$

$$T \longrightarrow \boxed{0}$$

The rule and its value

$$X(H) = 1, \quad X(T) = 0.$$

Once the outcome ω is known, the number $X(\omega)$ is determined.

$$\{X = 1\} = \{H\}, \quad P(X = 1) = P(\{H\}).$$

For every real t , $\{\omega \in \Omega : X(\omega) \leq t\}$ must be an event, so its probability is defined. This holds automatically on a finite sample space when every subset is an event.

A value of a random variable describes an event

Three independent fair coin tosses. Let X count the heads.

Outcomes	$X(\omega)$
TTT	0
HTT, THT, TTH	1
HHT, HTH, THH	2
HHH	3

The notation $\{X = 2\}$ means

$$\{\omega \in \Omega : X(\omega) = 2\}.$$

Here this event is

$$\{\text{HHT}, \text{HTH}, \text{THH}\}.$$

Each sequence has probability $1/8$.

Several outcomes can give the same value. Add the probabilities of those outcomes.

An indicator records whether an event occurs

For an event $A \subseteq \Omega$, define

Indicator of A

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

$$\{I_A = 1\} = A, \quad \{I_A = 0\} = A^c.$$

One fair die. Let $A = \{2, 4, 6\}$. List the values of I_A and find $P(I_A = 1)$.

The random variable records a fact about an outcome; it does not change the experiment.

Practice: translate a numerical condition into an event

Roll one fair die. Let $X(\omega)$ be its face value and set $Y(\omega) = I_{\{2,4,6\}}(\omega)$.

- 1 Describe $\{X \leq 3\}$ and $\{Y = 1\}$ as sets of outcomes.
 - 2 Find $P(X \leq 3)$, $P(Y = 1)$, and $P(X \leq 3, Y = 1)$.
 - 3 Are the two events independent?
-
-
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Begin with the outcomes, then compute their probabilities.

Events, variables, and values

Object	Meaning
$\omega \in \Omega$	An outcome of the experiment.
$X : \Omega \rightarrow \mathbb{R}$	A fixed rule assigning a number to each outcome.
$X(\omega)$	The value obtained at that outcome.
$\{X \leq t\}$	The event $\{\omega : X(\omega) \leq t\}$.
I_A	A variable recording whether A occurs.

Independence is a probability condition; it is not the same as disjointness.

Wednesday: from an outcome to a distribution

Recall the three-toss count

X = number of heads.

$$\Omega = \{HHH, HHT, \dots, TTT\}.$$

Today

- ① Collect probabilities into a PMF.
- ② Compare the PMF with its CDF.
- ③ Read probabilities with the correct endpoints.

Two different objects

HHT is an experimental outcome. The number 2 is the value of X at that outcome.

Check: name a different outcome with the same value of X .

Probability mass belongs to values

A discrete random variable takes values in a finite or countable set.

Probability mass function (PMF)

$$p_X(x) = P(X = x).$$

For its positive-mass values $S_X = \{x : p_X(x) > 0\}$,

$$p_X(x) \geq 0, \quad \sum_{x \in S_X} p_X(x) = 1.$$

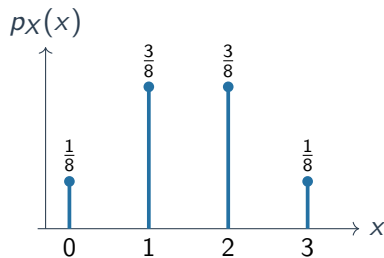
Why must the sum be one? Use the events $\{X = x\}$.

For the integer-valued models here, S_X is the support. Set the PMF to zero outside it.

The same distribution as a table and a PMF plot

For the number of heads in three independent fair tosses:

x	0	1	2	3
$p_X(x)$	$1/8$	$3/8$	$3/8$	$1/8$



Read probabilities from masses.

$$P(X \in \{1, 2\}) = p_X(1) + p_X(2).$$

The count is integer-valued: $P(X = 1.5) = 0$.

The CDF accumulates probability up to a threshold

Cumulative distribution function (CDF)

For every real t ,

$$F_X(t) = P(X \leq t).$$

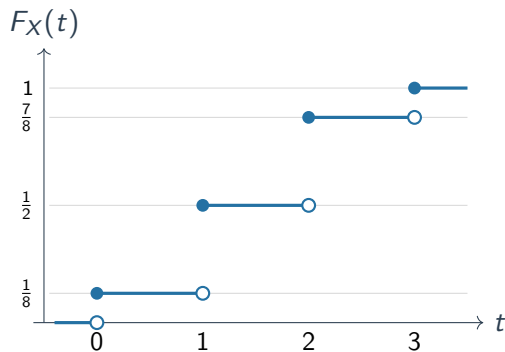
For a discrete X , this is $F_X(t) = \sum_{x \leq t} p_X(x)$.

For the three-toss count, fill in the cumulative values.

The threshold t may be noninteger even when X is integer-valued.

A discrete CDF is a step function

$$F_X(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{8}, & 0 \leq t < 1, \\ \frac{1}{2}, & 1 \leq t < 2, \\ \frac{7}{8}, & 2 \leq t < 3, \\ 1, & t \geq 3. \end{cases}$$



Filled dots give the function value. The jump at k is $P(X = k)$.

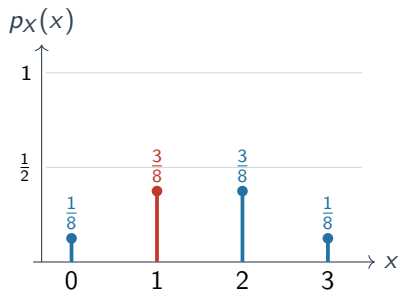
A CDF is nondecreasing and right-continuous, with limits 0 and 1 at $-\infty$ and $+\infty$.

PMF and CDF of the same random variable

X : number of heads in three independent fair tosses.

PMF: mass at a value

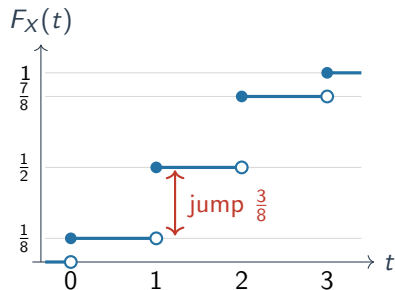
$$p_X(x) = P(X = x)$$



$$p_X(1) = \frac{3}{8}.$$

CDF: mass up to a threshold

$$F_X(t) = P(X \leq t)$$



$$F_X(1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}.$$

At 1, the PMF gives the CDF's jump: $\frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. Between possible values, the CDF stays constant.

Read the endpoints before subtracting CDF values

For real $a < b$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Split $\{X \leq b\}$ into $\{X \leq a\}$ and $\{a < X \leq b\}$.

For the three-toss count, compare:

$$P(X < 1), \quad P(X \leq 1), \quad P(1 < X \leq 2).$$

For integer-valued X and integer k : $P(X < k) = F_X(k - 1)$; $P(X > k) = 1 - F_X(k)$.

Practice: recover a CDF from three masses

x	-1	1	4
$p_X(x)$	$1/4$	$1/2$	$1/4$

- 1 Sketch $F_X(t)$; mark its value at each jump.
- 2 Find $P(X = 1)$, $P(X \leq 1)$, and $P(-1 < X \leq 4)$.

Point mass and accumulated mass

PMF

$$p_X(x) = P(X = x).$$

One value; nonnegative masses adding to one.

CDF

$$F_X(t) = P(X \leq t).$$

All values at or below a real threshold.

$$P(a < X \leq b) = F_X(b) - F_X(a), \quad a < b.$$

For our integer-valued examples, the jump at k is $p_X(k)$.

A CDF is nondecreasing and right-continuous. Its limits are 0 and 1 at $-\infty$ and $+\infty$.

Friday: a model for repeated trials

From the previous lessons

- Independence permits probabilities to factor.
- A random variable assigns a value to each outcome.
- A PMF assigns probabilities to those values.

Today

Count successes in a fixed number of independent trials.

one pattern's probability $\times$ number of patterns.

We will derive the binomial formula using independence and counting.

One Bernoulli variable and a count

For an event A_i with $P(A_i) = p$, define its indicator I_i .

Trial i

$$I_i = \begin{cases} 1, & \text{success,} \\ 0, & \text{failure.} \end{cases}$$

$$P(I_i = 1) = p, \quad P(I_i = 0) = 1 - p.$$

This is the Bernoulli law: $I_i \sim \text{Bern}(p)$.

Count in n trials

$$X = I_1 + \cdots + I_n.$$

$$X \in \{0, 1, \dots, n\}.$$

The event $\{I_i = 1\}$ is A_i . If the success events are mutually independent with common probability p , these are Bernoulli trials.

For the derivation, take $n \geq 1$ and $0 < p < 1$. The cases $p = 0$ and $p = 1$ give constant counts.

Independent trials: what the assumption means

Bernoulli trials

A fixed number $n \geq 1$ of trials, each with success probability p . The trials are mutually independent: the probability of every prescribed set of trial results factors into the corresponding individual probabilities.

$$P(I_1 = i_1, \dots, I_n = i_n) = \prod_{j=1}^n P(I_j = i_j), \quad i_j \in \{0, 1\}.$$

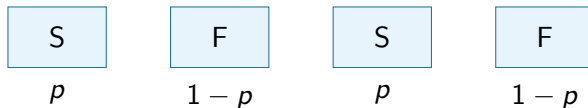
Check the full assumption

Pairwise independence alone does not justify this factorization.

For $0 < p < 1$, observing earlier results does not change a later trial's success probability.

First calculate the probability of one pattern

Four independent trials, each with success probability p .



Specified pattern: success, failure, success, failure.

For any specified pattern with k successes in n trials

$$\text{pattern probability} = p^k(1 - p)^{n-k}.$$

Then count the patterns

Let X count successes in n Bernoulli trials.

- ① Choose the k positions occupied by successes.
- ② Every such pattern has probability $p^k(1 - p)^{n-k}$.
- ③ Distinct patterns are disjoint outcomes, so add their probabilities.

Binomial distribution: $X \sim \text{Bin}(n, p)$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n.$$

The coefficient counts positions; the powers give one pattern's probability.

Exactly two successes or at least one?

A hypothetical randomized test

Four independent runs each reveal a particular defect with probability $1/4$. Let X count the runs that reveal it. Thus $X \sim \text{Bin}(4, 1/4)$.

Exactly two runs

At least one run

The complement is a single pattern: four failures.

Check the assumptions before choosing a binomial model

A binomial count has

a fixed number of trials, two outcomes per trial, a common p , and mutual independence.

Sampling without replacement

A bag has three red balls and two blue balls. Draw two balls without replacement. Let X count red balls. Is $X \sim \text{Bin}(2, 3/5)$?

A fixed trial count and two outcomes are not enough.

Practice: count heads in three fair tosses

Let X be the number of heads in three independent fair coin tosses.

- 1 Find $P(X = k)$ for $k = 0, 1, 2, 3$.
- 2 Explain the symmetry between k and $3 - k$.

This recovers Wednesday's table from a general formula.

Independence and counts: the essential distinctions

Question	What to check
Independent?	$P(A \cap B) = P(A) P(B)$.
Disjoint?	$A \cap B = \emptyset$; a condition on events, not factorization.
Several trials?	Pairwise independence alone does not justify a full pattern product.
Binomial count?	Fixed n , common p , and mutually independent Bernoulli trials.
At least one?	Consider $1 - P(\text{none})$.

State the model before substituting into a formula.

Optional: taking a complement preserves independence

Assume A and B are independent.

Claim

A and B^c are also independent.

Split A into the disjoint pieces $A \cap B$ and $A \cap B^c$.

Repeating this argument also gives independence of A^c and B^c .

Optional: pairwise independence does not imply mutual independence

Two independent fair coins. A : first H; B : second H; C : results agree.

Outcome	A	B	C
HH	1	1	1
HT	1	0	0
TH	0	1	0
TT	0	0	1

Each row has probability $1/4$.

Compare pairs with the triple.

Mutual independence of $A_1, \dots, A_n$

For every subset $J \subseteq \{1, \dots, n\}$ with $|J| \geq 2$,

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j).$$

References for this week

DeGroot and Schervish

Probability and Statistics, fourth edition.

- §2.2: independence.
- §§3.1, 3.3: discrete random variables and CDFs.
- §5.2: the binomial distribution.

Mor Harchol-Balter

Introduction to Probability for Computing, supplied 2024 version.

- Chapters 2–3: probability rules and discrete random variables.

Sheng Jiang

STAT 131 Notes, July 18, 2024: pp. 21–26 and 30–31.

Figures and exposition retain Antonio de Leon's earlier STAT 131 teaching style.